Biometric Authentication Mechanisms

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Abstract

Biometric credentials such as fingerprints and face recognition are widely used in modern authentication. Biometric sensors obtain a sample, calculate a *similarity score* and output accept if it is above a set threshold or reject otherwise. A large body of work focuses on improving the accuracy of biometric sensors. Other works study *authentication mechanisms* with one or more credentials, but treat biometric credentials as *atomic* (like passwords, either a perfect match or wrong); The state of the art for this prevalent authentication method is using a heuristic threshold.

We explore optimal-threshold authentication mechanisms with biometric credentials. Following prior work, we define a mechanism's success as the probability it accepts legitimate users and rejects attackers. We model user and attacker similarity scores as random distributions and symbolically derive optimal thresholds. Our optimal threshold reduces mechanism failure probability by 50-70% compared to the heuristic threshold, when biometric credentials are incorporated together or with atomic credentials.

We prove that integrating a biometric credential improves mechanism success even when an atomic credential has arbitrary low fault probabilities and a biometric credential provides only weak separation between users and attackers. We quantitatively demonstrate the improvement through experiments with real public biometric data, testing users against spoofing attempts. Our optimized threshold reduces mechanism failure probability by an order of magnitude compared to the heuristic threshold when a biometric credential is integrated with an atomic one.

These results imply that biometric sensors should stop defaulting to EER and allow setting their threshold according to the full authentication mechanism.

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1 Introduction

Authentication is a critical component of security, serving as the gatekeeper that verifies the identity of users and grants access to digital assets [4, 13, 22]. From the early days of the Internet, passwords were used to authenticate into the user account. With the rapid growth of online services, an increasing portion of daily activities, ranging from banking and commerce to communication, has transitioned to digital platforms. This expansion led companies to deploy advanced authentication mechanisms, such as multi-factor authentication (MFA) [3], which combines multiple independent credentials (e.g., passwords, tokens, or biometrics) for user's verification. However, password vulnerabilities persist: 59% of users in the U.S. reported leaks and 22% frequently forget their passwords, as of 2026 [11]. Consequently, biometric credentials are increasingly adopted, as they leverage unique physical characteristics of the user making them harder to steal, reuse, or forget.

Previous work (§2) has conducted extensive analysis of biometric credentials in isolation, primarily focusing on system accuracy and performance[9, 18]. Sarkar et al. [21] study biometric systems and introduce the Equal Error Rate (EER) as an operating point for biometric systems. However, this choice is made without a formal success function that justifies EER as optimal from a security perspective. Hong et al. [8] propose a fused biometric system in which the final decision is based on the matching scores of both face and fingerprint. While this approach captures dependencies between biometric sources through score fusion, it does not consider how one credential may influence the operating threshold of another, nor analyze the integration of other credential types within an authentication mechanism. Other works proposed and deployed various authentication solutions. These encompass mechanisms defined as Boolean criterion over credential possession [3, 6, 13, 16], where each credential is treated in an all or nothing manner. However, such settings are not directly applicable to our framework, as they assume credentials produce binary outcomes and do not account for the score outputs and threshold adaptation of biometric credentials.

In this work, we study biometric credentials in isolation as well as their integration with other types of credentials within an authentication mechanism. Our goal is to analyze how the presence of heterogeneous credentials can influence and determine the biometric decision threshold. First, we define the authentication model (§3) built on a set of biometric and atomic credentials, requires an indivisible, exact match to grant access. The mechanism interacts with 2 players: a user and an attacker, trying to identify the user based on the provided credentials by the player. We follow the definitions given by Eyal [6], where the possesion of each credential can be in one of the following states: safe, loss, leak and theft. Then, we associate a probability with each one of the states, and define the mechanim's success as the probability it accepts legitimate users and rejects attackers. Subsequently, we extend this framework to biometric credentials, which are evaluated by a mechanism that outputs a confidence score for both players. This score characterizes the geniune and imposter distributions, which are used to establish two error rates: false acceptance rate (FAR)-The probability that the system incorrectly accepts an attacker as a legitimate user, and false rejection rate (FRR)-The probability that the system incorrectly rejects a legitimate user.

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We start with analyzing the effect of a single biometirc credential on the mechnaism's success (§4). We examine distribution functions that approximate a biometric credential and formalize an optimal decision threshold. We symbolically calculate the optimal threshold and find that it is different than the commonly used one, where the FAR equals FRR, known as the equal error rate (EER) [21].

As each credential can influence the mechanism's success probability differently, we examine a more complex and realistic case, where a mechanism is comprised of multiple hetrogenous credentials (§5). Our analysis reveals that the optimal threshold for a mechanism comprised of multiple biometric credentials is not necessarily the same as the optimal threshold if used individually (§5.1). Using a numeircal optimization algorithm, we observe a reduction of over 66% in the failure rate of the success function, when comparing the EER and the optimal threshold.

We find that the optimal threshold would differ, depending on how a user integrates the biometric credential. We link the values of the atomic credential to the influence they have on the decision threshold of the biometric credential. This relation shows that high safe values and low leak values of the atomic credential shift the threshold towards the lower boundary of the biometric credential, making the mechanism's success mainly dependent on the atomic credential, and vice versa.

This observation, suggests that the biometric credential is nearly degenerated. Then, we characterize the conditions under which integrating a biometric credential into a mechanism composed of a single atomic credential, enhances its success (§6). We prove that when these conditions are not satisfied, the integration is suboptimal. Furthermore, we show that the improvement is consistent across all *asymptotic separabile* biometric credentials which is a necessary and sufficient condition, provided that the atomic credential has non-zero probabilities for the states of safe and either loss or leak. This finding has immediate and practical implications for the design and improvement of authentication mechanisms, indicating which biometric credentials can enhance the mechanism's success. To provide a more practical criterion, we introduce *asymptotic dominancy*, a sufficient condition for the enhancement of the mechanism's success. This condition implies that the attacker's distribution is dominated by the user's distribution as the matching score s goes to the distribution limits.

To support our theoretical analysis, we conduct empirical experiments using real biometric data and SourceAfis, an open source algorithm for fingerprint verification [24] (§7). We use a dataset of fingerprints from multiple users and evaluate matching scores by comparing samples of one user against those of other users. In this setting, we observe high seperability between the two players, as the algorithm effectively extracts minutiae and distinguishes between different individuals. Consequently, when different users attempt to authenticate as one another, no significant improvement is observed when comparing the mechanism's success under Equal Error Rate (EER) threshold to the optimal threshold. However, more sophisticated attack scenarios involve attempts to impersonate a legitimate user. To model this, we conduct a spoofing attack simulation using the LivDet database [7, 14, 17, 25], which includes both genuine fingerprint scans and corresponding spoofing attempts. These spoof samples produce matching scores that closely resemble those of the legitimate user. Using SourceAFIS, we observe that integrating the biometric credential into a mechanism comprised of atomic credential, reduces the mechanism's failure probability by 56% compared to the probability with EER threshold. Furthermore, the results show a 50% reduction in the mechanism's failure probability compared to a mechanism based solely on atomic credential, demonstrating the benefit of integrating the biometric credential.

In summary (§8) our contributions are both theoretical and practical: (1) Formalization of the mechanism's success function composed of biometric credentials and heterogeneous; (2) Symbolic analysis of the optimal threshold for biometric and heterogeneous authentication mechanisms. (3) Characterization of the conditions under which biometric credential integration enhances the mechnaims's success; (4) Proof that any biometric credential, under a specified condition, always enhances the mechnaims's success. (5) Empirical analysis using real biometric data, demonstrating theoretical analysis. (6) Direct implications to how fingerprint readers should work in practice for immediate security improvements.

2 Related Work

Authentication remains an active and prominent research area, with prior work broadly categorized into two main aspects: credentials and mechanisms. Credentials form the core of the authentication process, commonly manifested as passwords [15], OTPs [19], physical devices [23] or biometric features such as fingerprints, facial recognition or iris [10]. An authentication mechanism is a decision making protocol that evaluates a user's legitimacy based on one or more credentials. It defines how these credentials are evaluated and combined to produce a final decision. Examples include multi-factor authentication (MFA) [3], where each credential is typically treated as a binary factor.

Recent work has explored more sophisticated mechanism designs. Maram et al. [13] study interactive authentication mechanisms, where the user or attacker interacts iteratively with the authentication mechanism. Mouallem et al. [16] compare mechanisms' security and aim to identify mechanisms with maximal security when composed of a large number of homogeneous credentials. Eyal [6] analyzes mechanisms modeled as Boolean functions of the credentials. However, these works are not directly applicable to our setting, as they assume credentials can be modeled as binary factors. Although biometric systems typically produce binary decisions, these decisions are derived from continuous scores, and reducing them to Boolean values obscures essential information.

Complementary research focuses specifically on biometric credentials, largely in isolation from authentication mechanisms. Extensive work has evaluated the performance and accuracy of biometric systems [5, 7, 9, 12, 14, 17, 25]. Prabhakar et al. [18] examine the tradeoff between two key metrics defining the accuracy of biometric systems, namely False Acceptance Rate (FAR) and False Rejection Rate (FRR). Sarkar et al. [21] study biometric credentials in isolation. They introduce the Equal Error Rate (EER), the point where FAR and FRR are equal, and identify it as an operating point of the biometric system. Hong et al. [8] integrate fingerprint and face recognition to design a fused biometric system that combines the matching scores of both modalities to make a final decision.

Critically, none of these works define a rigorous success criterion for biometric systems, nor do they analyze the integration of biometric credentials with other credential types. Moreover, they assume a fixed operating threshold for the biometric system and do not explore how the presence of additional credentials can influence the optimal decision threshold of the biometric system, which is a key aspect of our work.

3 Authentication Model

An authentication mechanism M is built on a set of credentials $\{c_1, c_2, \ldots, c_n\}$ and two players: a user U and an attacker A. Both players, try to satisfy the system with their own set of credentials $\{c_1, c_2, \ldots, c_n\}$ in order to reach the asset. Each credential c_i can be either discrete or continuously distributed. In this paper, we will focus on continuously distributed credentials. We will formalize the systems success and identify the optimal operating point for both single and multiple credentials.

3.1 Model Details

We follow the definitions given by Eyal [6]. A wallet w is defined by predicate of availability of N keys. Each key can be in one of the following states. Safe: only the user has access to the key. Loss: neither the user nor the attacker has access to the key. Leak: both the user and the attacker have access to the key. Theft: only the attacker has access to the key.

We define the probability associated with each state as follows: P_{safe}, P_{loss}, P_{leak} and P_{theft} corresponding respectively to the states of safe, loss, leak, and theft. Furthermore, the confidence associated with different decisions may be characterized by the genuine distribution and the impostor distribution, which are used to establish two error rates: false acceptance rate (FAR), which is defined as the probability of an impostor being accepted as a genuine individual and false rejection rate (FRR), which is defined as the probability of a genuine individual being rejected as an impostor.

A wallet w is comprised of a set of credentials $\{c_1, c_2, \ldots, c_n\}$ clasified as one of previously defined states.

Denoted S, i.e., $\{c_1, c_2, \ldots, c_n\} \in S = \{\text{safe}, \text{loss}, \text{leak}, \text{theft}\}$ the state of each credential. A scenario σ is defined as a vector of states of each credential in the wallet. Denote σ_i the state of credential c_i in scenario σ. An availability vector represents the availablity of all credentials to a user U or an attacker A. Denote by σ^U,σ^A the availability vector of the user and the attacker respectively.

In the following sections we will represent all biometric credentials as the probability function of the matching score s denoted as $P_U[s]$. We follow the definitions given by Jain et al. [10] and generalize it to all biometric credentials. If the stored reference template of the user U is represented by X_U and the acquired input for recognition is represented by X_Q then:

H_0 input X_Q does not come from the same person as the reference template X_U.

H_1 input X_Q comes from the same person as the reference template X_U.

D_O, D_1 are the imposter and geniuine persons trying to authenticate into the account. The matching score s, is usually a single number, that quantifies the similarity between an input X_Q and and the reference store template in the database X_U [18]. If the matching score $s(X_Q, X_U)$ is above a threshold t then decide D_1 else, decide D_0.

4 Single Credential

For a wallet w comprised of a single credential c_1, $w^{\text{single}}(c_1) = c_1$ we have built FAR vs FRR curves for Uniform and Quadratic probabilistic distribution functions and defined each one of the states as follows:

$$P_{\text{loss}} = \text{FRR} \cdot (1 - \text{FAR})$$

$$P_{\text{leak}} = \text{FAR} \cdot (1 - \text{FRR})$$

$$P_{\text{theft}} = \text{FRR} \cdot \text{FAR}$$

$$P_{\text{safe}} = 1 - \text{P}_{\text{loss}} - \text{P}_{\text{leak}} - \text{P}_{\text{theft}}$$

The probability for success of the wallet w, is defined as the probability of users successfully defend their wallets. In a single credential wallet, this is simply the probability that the key is neither theft, loss or leaked. i.e.,

$$P_{\text{success}}(w^{\text{single}}) = P_{\text{safe}} = 1 - \text{FAR} - \text{FRR} + \text{FAR} \cdot \text{FRR} \tag{1}$$

Until now, it has been common practice to select the operating point at the Equal Error Rate (EER), where FAR = FRR following Sarkar's claim: "The operating point is that point where the FAR is equal to the FRR and known as Equal Error Rate (EER)" [21]. Our findings indicate that a different operating point can yield a higher P_{safe}, thereby enhancing the system's overall security. Furthermore, We found that the operating point depends on the variance of the PDFs. We will further explore this in the following sections for different distributions.

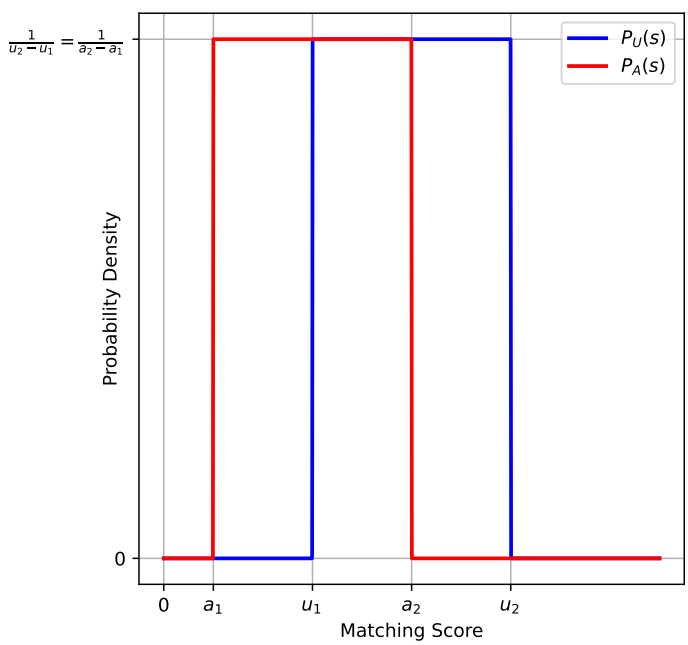
4.1 Uniformly Distributed Credential

As a first-order approximation, we model the biometric credential using a uniform distribution. We defined two probability distribution functions for the user and the attacker, P_U and P_A respectively.

$$P_U(s) = \frac{1}{u_2 - u_1} \quad \text{for } u_1 \leq s \leq u_2$$

$$P_A(s) = \frac{1}{a_2 - a_1} \quad \text{for } a_1 \leq s \leq a_2$$

Where u_1, u_2 are the user PDFs' bounds and a_1, a_2 are the attacker PDFs' bounds.



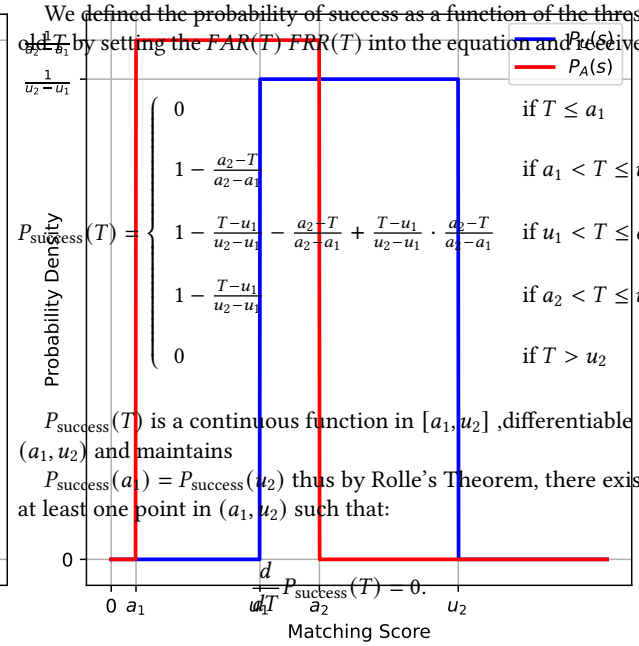


Figure 1: Symmetric and Asymmetric Uniform PDFs

We assume that $a_1 < u_1$ and $a_2 < u_2$ as the x axis represents the matching score s and higher values of s corresponds to the user D_1 as mentioned in the model details. Furthemore, we assume there is an overlaping region between the distributions; otherwise the scenario reduces to a trivial case. We also assume that the user and attacker PDFs are do not fully contain one another, as such cases would render them non-separable. Figure 1a illustrates the case where the user and attacker PDF's are symmetric, while Figure 1b illustrates asymmetric PDFs both as function of the matching score s. The probability of success for a single credential wallet is defined as:

$$P_{\text{success}}(w^{\text{single}}) = P_{\text{safe}} = 1 - \text{P}_{\text{loss}} - \text{P}_{\text{leak}} - \text{P}_{\text{theft}} = 1 - \text{FRR} - \text{FAR} + \text{FRR} \cdot \text{FAR}$$

Let T denote the threshold. Then the False Acceptance Rate (FAR) and False Rejection Rate (FRR) are defined as:

$$\text{FAR}(T) = \int_T^{a_2} \frac{1}{a_2 - a_1} ds = \begin{cases} 1 & \text{if } T \le a_1 \\ \frac{a_2 - T}{a_2 - a_1} & \text{if } a_1 < T \le a_2 \\ 0 & \text{if } T > a_2 \end{cases}$$

$$\text{FRR}(T) = \int_{u_1}^{T} \frac{1}{u_2 - u_1} ds = \begin{cases} 0 & \text{if } T \le u_1 \\ \frac{T - u_1}{u_2 - u_1} & \text{if } u_1 < T \le u_2 \\ 1 & \text{if } T > u_2 \end{cases}$$

We defined the probability of success as a function of the threshold T by setting the $FAR(T)$ $FRR(T)$ into the equation and received:

$$P_{\text{success}}(T) = \begin{cases} 0 & \text{if } T \le a_1 \\ 1 - \frac{a_2 - T}{a_2 - a_1} & \text{if } a_1 < T \le u_1 \\ 1 - \frac{T - u_1}{u_2 - u_1} - \frac{a_2 - T}{a_2 - a_1} + \frac{T - u_1}{u_2 - u_1} \cdot \frac{a_2 - T}{a_2 - a_1} & \text{if } u_1 < T \le a_2 \\ 1 - \frac{T - u_1}{u_2 - u_1} & \text{if } a_2 < T \le u_2 \\ 0 & \text{if } T > u_2 \end{cases}$$

$P_{\text{success}}(T)$ is a continuous function in $[a_1, u_2]$,differentiable in (a_1, u_2) and maintains

$P_{\text{success}}(a_1) = P_{\text{success}}(u_2)$ thus by Rolle's Theorem, there exists at least one point in (a_1, u_2) such that:

$$\frac{d}{dT} P_{\text{success}}(T) = 0.$$

The horizontal tangent point is:

$$T = \frac{a_1 + u_2}{2} \quad \text{for } u_1 \le T \le a_2$$

$$\frac{d}{dt} P_{\text{success}}(T) > 0, \text{For every } T \in (u_1, \frac{a_1 + u_2}{2})$$

$$\frac{d}{dt} P_{\text{success}}(T) < 0, \text{For every } T \in (\frac{a_1 + u_2}{2}, a_2)$$

Thus, by the First Derivative Test the maximum point is at $T = \frac{a_1 + u_2}{2}$.

$$P_{\text{success}}(T_{\text{opt}}) = \frac{(a_1 - u_2)^2}{4(a_1 - a_2)(u_1 - u_2)}$$

We calculated the EER point and showed it is not equal to the optimal point:

$$T_{\text{EER}} = \frac{a_1 \cdot u_1 - a_2 \cdot u_2}{a_1 - a_2 + u_1 - u_2} \neq \frac{a_1 + u_2}{2}$$

The maximum point represnents the optimal threshold of the wallet. By selecting the threshold value, we influnce the wallet's security sensetivity by adjusting the probability associated with each state in S. The optimal point indicates that increasing the offset between the two uniform distributions is likely to improve the wallet's success. This optimal point suggest an asymmetry between the user's and the attacker's distribution intervals. Figure 2 illustrates the probability of success for a single credential wallet depends on the threshold T.

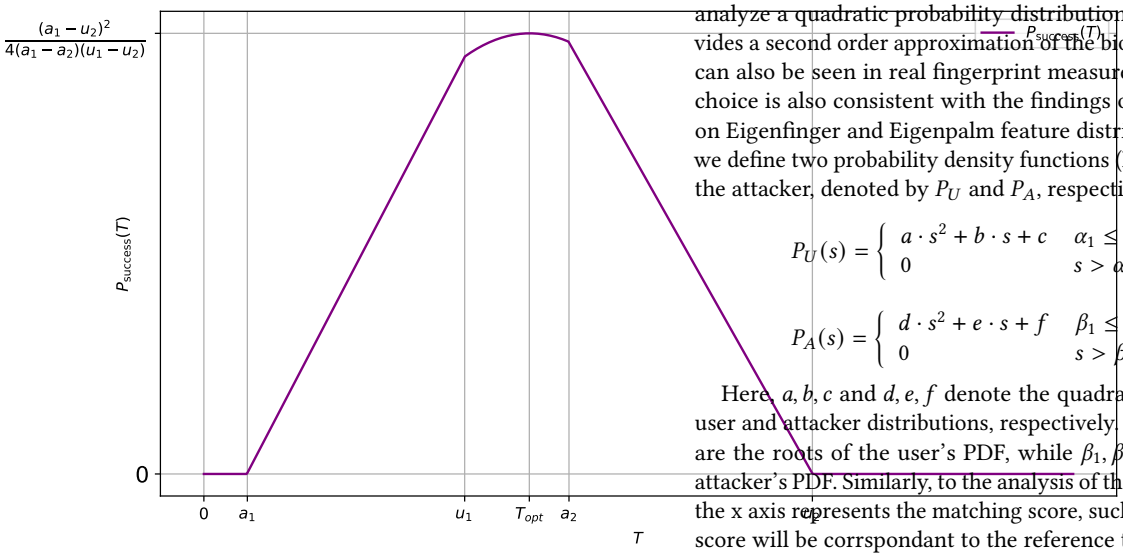


Figure 2: Probability of success for a single credential wallet asymmetrically distributed

We can define the FAR and FRR at the optimal point as follows:

$$\mathrm{FAR}(T_{\mathrm{opt}}) = \frac{2a_2 - a_1 - u_2}{2(a_2 - a_1)}$$

$$\mathrm{FRR}(T_{\mathrm{opt}}) = \frac{a_1 + u_2 - 2u_1}{2(u_2 - u_1)}$$

Figure 3 illustrated the FAR vs FRR curve.

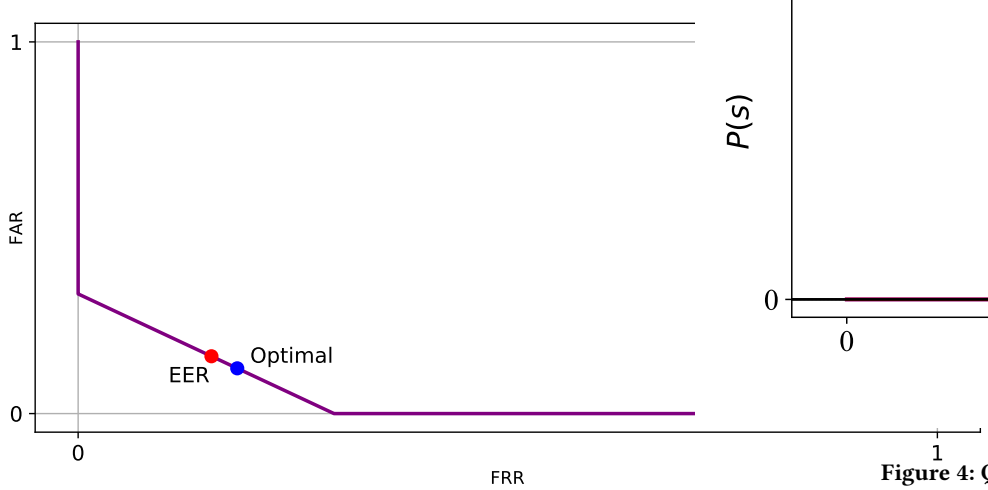


Figure 3: FAR vs FRR curve

4.2 Quadratic Distributed Credential

Our goal is to model credentials whose distributions closely resemble those observed in real measurement data. To this end, we analyze a quadratic probability distribution function, which provides a second order approximation of the biometric credential that can also be seen in real fingerprint measurement data (§7). This choice is also consistent with the findings of Slobodan et al. [20] on Eigenfinger and Eigenpalm feature distributions. Accordingly, we define two probability density functions (PDFs) for the user and the attacker, denoted by P_U and P_A, respectively.

$$P_U(s) = \begin{cases} a \cdot s^2 + b \cdot s + c & \alpha_1 \le s \le \alpha_2 \\ 0 & s > \alpha_2 \vee s < \alpha_1 \end{cases}$$

$$P_A(s) = \begin{cases} d \cdot s^2 + e \cdot s + f & \beta_1 \le s \le \beta_2 \\ 0 & s > \beta_2 \vee s < \beta_1 \end{cases}$$

Here, a, b, c and d, e, f denote the quadratic coefficients of the user and attacker distributions, respectively. The parameters α_1, α_2 are the roots of the user's PDF, while β_1, β_2 are the roots of the attacker's PDF. Similarly, to the analysis of the uniform distribution, the x axis represents the matching score, such that higher values of score will be corrspondant to the reference template stored in the database X_U. If $\beta_2 < \alpha_1$, the two distributions do not overlap, and the problem becomes trivial. Conversely, if $\alpha_1 > \beta_1$ and $\beta_2 < \alpha_2$, the attacker's distribution is fully contained within the user's, resulting in a non-separable scenario. We therefore focus on the intermediate and most meaningful configuration, where

$$\beta_1 \le \alpha_1 \le \beta_2 \le \alpha_2,$$

as illustrated in Figure 4.

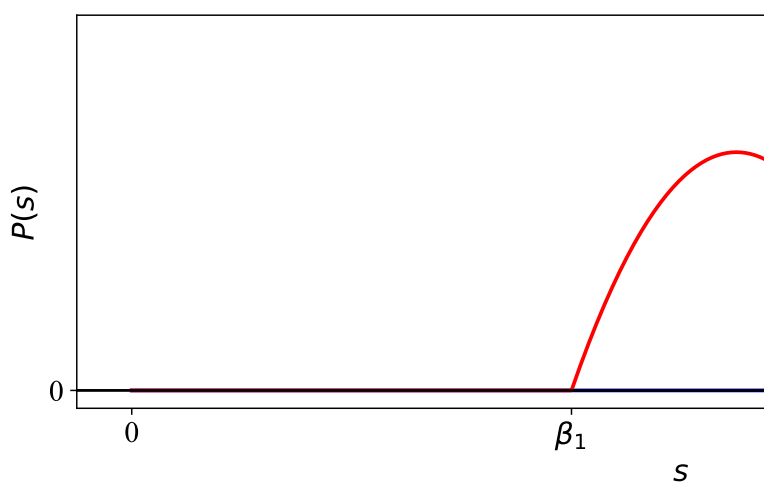


Figure 4: Quadratic PDFs

To reduce the degrees of freedom of the quadratic functions while maintaining a symbolic analysis, we express the coefficients directly in terms of their roots. Under the additional assumption that the integral of each PDF over its roots interval equals 1, the coefficients become fully determined. Accordingly, we define the coefficients as follows.

$$\alpha_1 \cdot \alpha_2 = \frac{c}{a} \rightarrow a = \frac{c}{\alpha_1 \cdot \alpha_2}$$

$$\alpha_1 + \alpha_2 = -\frac{b}{a} \rightarrow b = -\frac{c \cdot (\alpha_1 + \alpha_2)}{\alpha_1 \cdot \alpha_2}$$

$$\beta_1 \cdot \beta_2 = \frac{f}{d} \rightarrow d = \frac{f}{\beta_1 \cdot \beta_2}$$

$$\beta_1 + \beta_2 = -\frac{e}{d} \rightarrow e = -\frac{f \cdot (\beta_1 + \beta_2)}{\beta_1 \cdot \beta_2}$$

$$\int_{\alpha_1}^{\alpha_2} (a \cdot x^2 + b \cdot x + c) dx = 1 \Rightarrow c = \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3}$$

$$\int_{\beta_1}^{\beta_2} (d \cdot x^2 + e \cdot x + f) dx = 1 \Rightarrow f = \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3}$$

Therefore, we can write the quadratic PDFs as:

$$P_U(s) = \begin{cases} \frac{6}{(\alpha_1 - \alpha_2)^2} \cdot s^2 - \frac{6 \cdot (\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3} \cdot x + \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3} & \alpha_1 \le x \le \alpha_2 \\ 0 & x > \alpha_2 \vee x < \alpha_1 \end{cases}$$

$$P_A(s) = \begin{cases} \frac{6}{(\beta_1 - \beta_2)^2} \cdot x^2 - \frac{6 \cdot (\beta_1 + \beta_2)}{(\beta_1 - \beta_2)^3} \cdot x + \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3} & \beta_1 \le x \le \beta_2 \\ 0 & x > \beta_2 \vee x < \beta_1 \end{cases}$$

We will find FAR and FRR as a function of the threshold T and define the probability of success based on the single credential wallet as shown in (§4.1):

$$\mathrm{FAR}(T) = \int_T^{\beta_2} P_A(s) ds = \begin{cases} 0 & T \le \beta_1 \\ \frac{6\left(\frac{\beta_2^3}{3} - \frac{T^3}{3}\right)}{(\beta_1 - \beta_2)^3} - \frac{6\left(\frac{\beta_2^2}{2} - \frac{T^2}{2}\right)(\beta_1 + \beta_2)}{(\beta_1 - \beta_2)^3} + \frac{6\beta_1\beta_2(\beta_2 - T)}{(\beta_1 - \beta_2)^3} & \beta_1 < T < \beta_2 \\ 1 & T > \beta_2 \end{cases}$$

$$\mathrm{FRR}(T) = \int_{\alpha_1}^{T} P_U(s) ds = \begin{cases} 1 & T \le \alpha_1 \\ \frac{6\left(\frac{T^3}{3} - \frac{\alpha_1^3}{3}\right)}{(\alpha_1 - \alpha_2)^3} - \frac{6\left(\frac{T^2}{2} - \frac{\alpha_1^2}{2}\right)(\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3} + \frac{6\alpha_1\alpha_2(T - \alpha_1)}{(\alpha_1 - \alpha_2)^3} & \alpha_1 < T \le \alpha_2 \\ 0 & T > \alpha_2 \end{cases}$$

$$P_{\text{success}} = \begin{cases} 0 & T < \beta_1 \vee T > \alpha_2 \\ \frac{(T-\beta_1)^2(2T+\beta_1-3\beta_2)}{(\beta_1-\beta_2)^3} & \beta_1 \le T \le \alpha_1 \\ -\frac{(T-\alpha_2)^2(T-\beta_1)^2(2T-3\alpha_1+\alpha_2)(2T+\beta_1-3\beta_2)}{(\alpha_1-\alpha_2)^3(\beta_1-\beta_2)^3} & \alpha_1 \le T \le \beta_2 \\ -\frac{(T-\alpha_2)^2(2T-3\alpha_1+\alpha_2)}{(\alpha_1-\alpha_2)^3} & \beta_2 \le T \le \alpha_2 \end{cases}$$

We differentiate $P_{\text{success}}(T)$ and compute its critical points in each interval. By comparing the stationary values obtained in all valid regions, we identify the global maximum, which corresponds to the optimal threshold T_{opt}.

Before writing the piecewise derivative, we define the polynomial coefficients:

$$q = 4,$$

$$m = -(5\alpha_1 + \alpha_2 + \beta_1 + 5\beta_2),$$

$$n = \alpha_1(3\alpha_2 + \beta_1 + 6\beta_2) - \alpha_2^2 + \alpha_2\beta_2 - \beta_1^2 + 3\beta_1\beta_2,$$

$$\ell = \alpha_1\left(\beta_1^2 - 3\beta_2(\alpha_2 + \beta_1)\right) + \alpha_2^2\beta_2.$$

With these coefficients, the derivative of $P_{\text{success}}(T)$ is given by the following piecewise expression:

$$\frac{dP_{\text{success}}}{dT} = \begin{cases} 0, & T < \beta_1 \ \vee \ T > \alpha_2, \\ \frac{6(T-\beta_1)(T-\beta_2)}{(\beta_1-\beta_2)^3}, & \beta_1 \le T \le \alpha_1, \\ -\frac{6(T-\alpha_2)(T-\beta_1)\left(qT^3 + mT^2 + nT + \ell\right)}{(\alpha_1-\alpha_2)^3(\beta_1-\beta_2)^3}, & \alpha_1 \le T \le \beta_2, \\ -\frac{6(T-\alpha_1)(T-\alpha_2)}{(\alpha_1-\alpha_2)^3}, & \beta_2 \le T \le \alpha_2. \end{cases}$$

In the second interval, the derivative has a single critical point at $T = \beta_1$. Since the derivative is strictly positive for all $T > \beta_1$, this point corresponds to a local minimum rather than a maximum. Similarly, in the fourth interval, the only critical point occurs at $T = \alpha_2$, and the derivative is strictly negative for all $T < \alpha_2$. Therefore, this interval also contains no maximum either.

We therefore turn to the third interval. In this region, the derivative is the product of a cubic polynomial and two linear factors. The linear factors vanish at $T = \beta_1$ and $T = \alpha_2$, both of which lie outside the third interval. Consequently, the existence of critical points within this interval depends solely on the cubic term. Solving this cubic ,reveals exactly one real root with the remaining two roots being complex. The closed form of the cubic solution can not be simplified further more than a complex expression. Thus, we denote the solution as T_{cubic}.

By examining the sign of the derivative on both sides of the critical point T_{cubic}, we verify that it constitutes a global maximum of $P_{\text{safe}}(T)$. We denote this maximizing point by T_{opt}.

To demonstrate that this optimal threshold differs from the Equal Error Rate (EER) threshold, we computed the EER point separately as follows:

$$T_{\text{EER}} = \frac{\alpha_1\beta_1 - \alpha_2\beta_2}{\alpha_1 - \alpha_2 + \beta_1 - \beta_2} \neq T_{\text{opt}}$$

We illustrate several user and attacker distributions and present their corresponding probability of success curves, expressed as functions of the threshold T in Figure 5 and Figure 6 respectively.

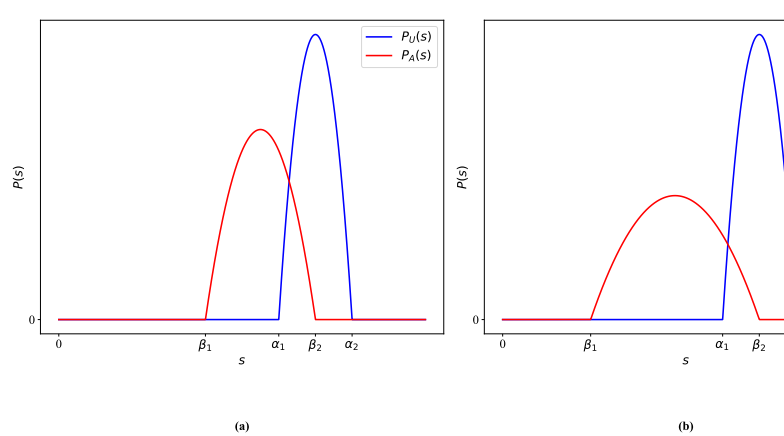


Figure 5: User and Attacker Quadratically Distributed

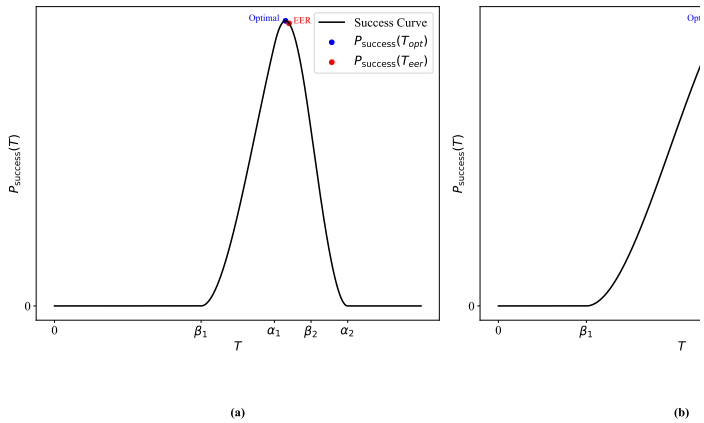


Figure 6: Probability of success for a single credential wallet quadratically distributed

Moreover, we illustrate the FAR–FRR curves for each pair of user and attacker distributions, and mark both the optimal point and the EER point, as shown in Figure 7.

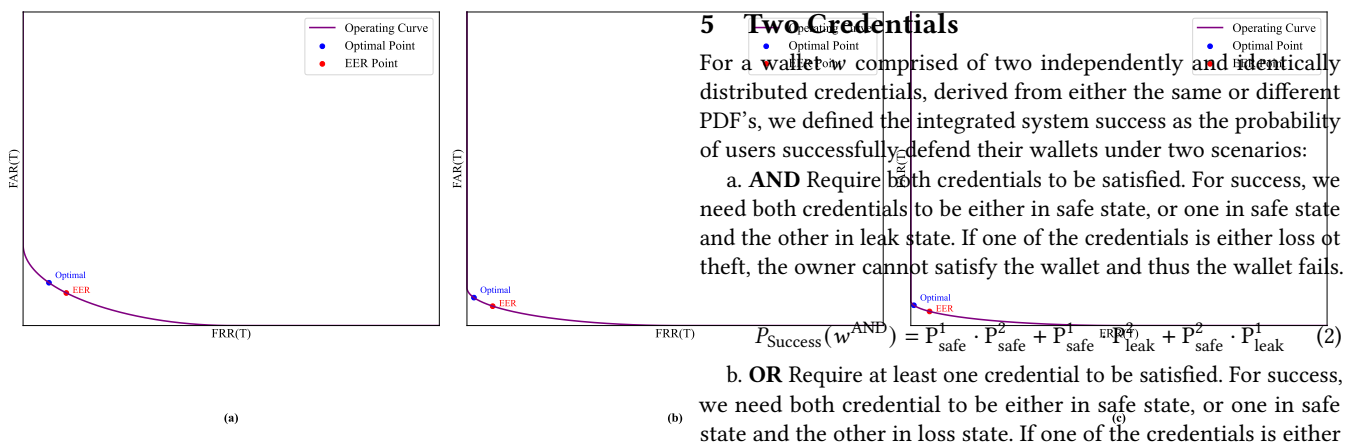


Figure 7: FAR vs FRR curve quadratically distributed

To quantify the effect of the variance on the deviation between the optimal operating point T_{opt} and the equal error rate (EER) point T_{EER}, we numerically computed this gap as a function of the attacker's variance. Decreasing the variance of the attacker achieved by reducing the distance between its roots, corresponds to reduced separability between the two distributions. The simulations reveal that as the variance increases, the gap $|T_{\text{opt}} - T_{\text{EER}}|$ grows accordingly, until the operating threshold moves into a regime where it effectively minimizes only one of the error rates, either FAR or FRR, driving the other to zero. This phenomenon is shown in Figure 8. These findings indicate that higher attacker variance enlarges the potential security advantage gained by operating at T_{opt} rather than at the EER point.

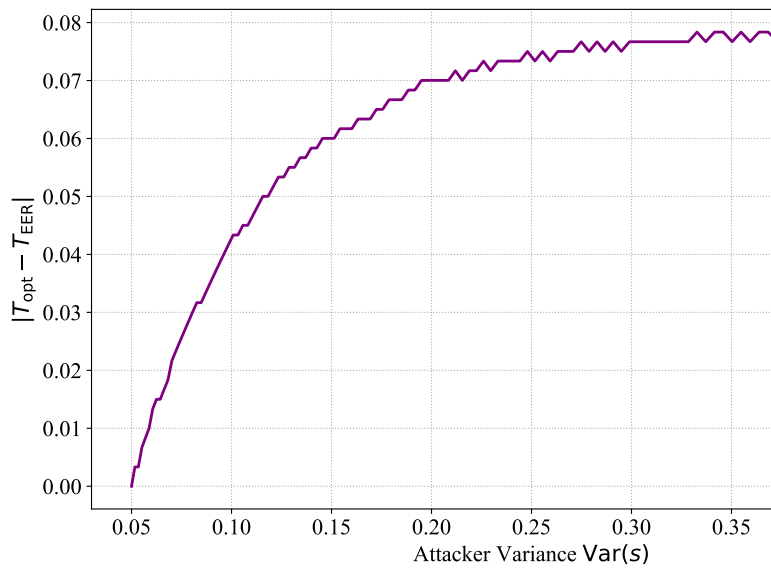


Figure 8: Gap between optimal point and EER point vs Variance of the attacker quadratically distributed

5 Two Credentials

For a wallet w comprised of two independently and identically distributed credentials, derived from either the same or different PDF's, we defined the integrated system success as the probability of users successfully defend their wallets under two scenarios:

a. **AND** Require both credentials to be satisfied. For success, we need both credentials to be either in safe state, or one in safe state and the other in leak state. If one of the credentials is either loss ot theft, the owner cannot satisfy the wallet and thus the wallet fails.

$$P_{\text{Success}}(w^{\text{AND}}) = \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{safe}} + \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{leak}} + \text{P}^2_{\text{safe}} \cdot \text{P}^1_{\text{leak}} \quad (2)$$

b. **OR** Require at least one credential to be satisfied. For success, we need both credential to be either in safe state, or one in safe state and the other in loss state. If one of the credentials is either leak ot theft, the attacker can satisfy the wallet and thus the wallet fails.

$$P_{\text{Success}}(w^{\text{OR}}) = \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{safe}} + \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{loss}} + \text{P}^2_{\text{safe}} \cdot \text{P}^1_{\text{loss}} \quad (3)$$

For each wallet type, we analyzed the case of two independently and identically distributed credentials in 2 cases:

a. A mechanism comprised of 2 biometric credentials.

b. A mechanism comprised of 1 discrete credential and 1 biometric credential.

We derived the optimal operating point for both **AND** and **OR** wallets by maximizing the respective success probabilities with respect to the thresholds of each credential.

5.1 2 Biometric Credentials

To analyze the case of a wallet w comprised of two biometric credentials, we conduct a numerical study over several probability

distribution functions. For each distribution pair, we derive the optimal operating point for both **AND** and **OR** wallets by maximizing the respective success probabilities with respect to the thresholds of each credential, using a LBFGS optimization algorithm. We additionally compare the optimal operating point to the EER point for each distribution type, thereby characterizing the conditions under which the **AND** or **OR** wallet yields superior performance. We started by analyzing the case of Gaussian distributions. We defined the user and attacker distributions as $P_U(s) = \mathcal{N}(\mu_u, \sigma_u^2)$ and $P_A(s) = \mathcal{N}(\mu_a, \sigma_a^2)$ respectively. We set for the first credential $\mu_u = 0.8$, $\sigma_u = 0.12$, $\mu_a = 0.3$ and $\sigma_a = 1$ and for the second credential $\mu_u = 1.0$, $\sigma_u = 0.2$, $\mu_a = 0.4$ and $\sigma_a = 0.18$ as shown in Figure 9.

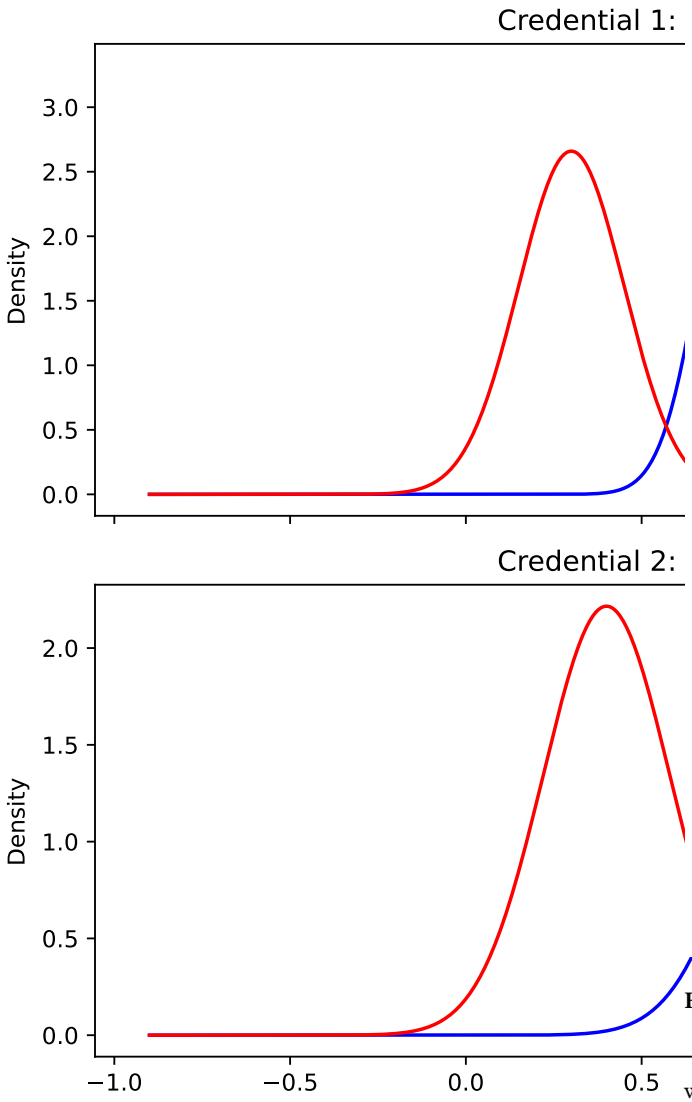


Figure 9: Gaussian distributed credentials

The optimal operating point for both wallet types was found to differ from the EER point, with the optimal point yielding a higher success probability than the EER point. Moreover, we found out that optimal point of both **AND** and **OR** wallets shifts from the individual optimum point of the credential. Hence, integrating the two credentials in a wallet improves performance over using them individually as shown in Figure 10.

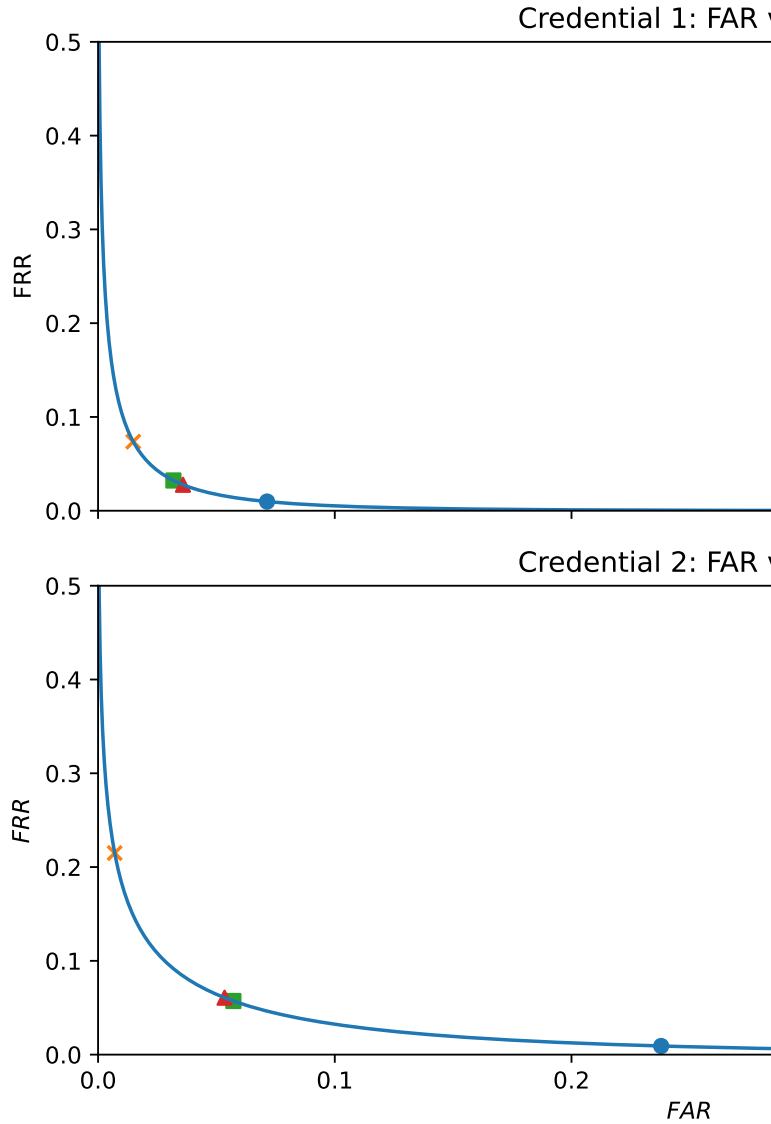
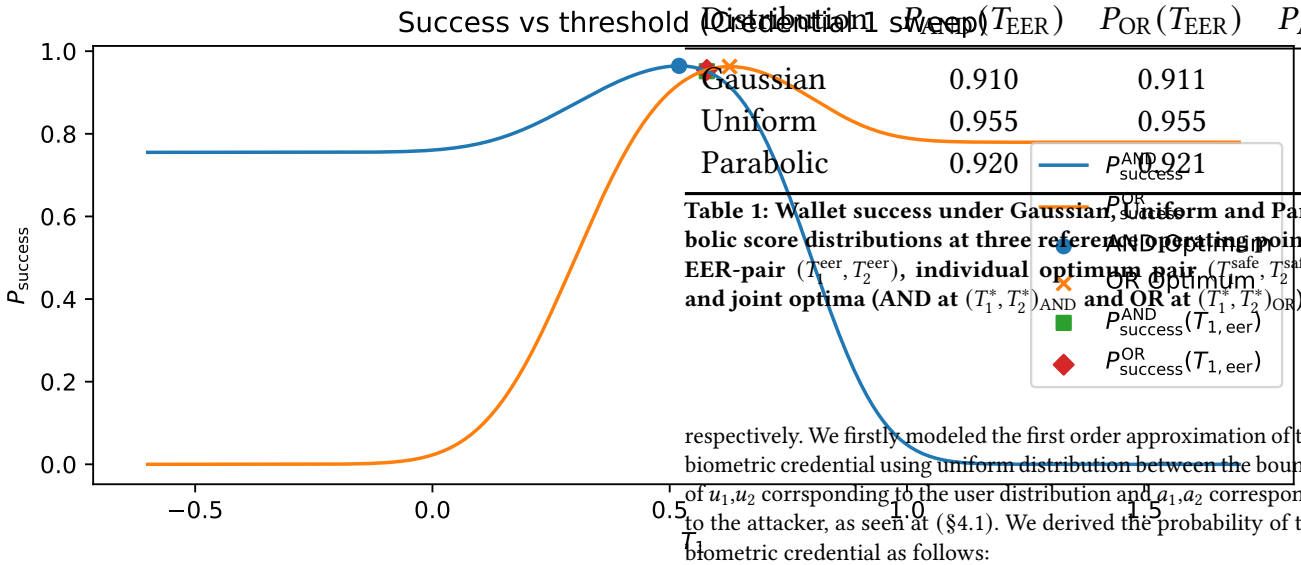


Figure 10: FAR vs FRR curves for the two Gaussian credentials

We illustrated the success probability of both **AND** and **OR** wallets. We fixed a credential to its optimum found out using the LBFGS algorithm and showed how the success probability varies as a function of the thresholds of the other credential. The results are shown in Figure 11. In conclusion, all the results are summarized in Table 1.



Distribution	$P_{\text{AND}}(T_{\text{EER}})$	$P_{\text{OR}}(T_{\text{EER}})$	$P_{\text{AND}}(T_1^{\text{safe}})$
Gaussian	0.910	0.911	0.911
Uniform	0.955	0.955	0.916
Parabolic	0.920	0.921	0.919

Table 1: Wallet success under Gaussian, uniform and Parabolic score distributions at three reference operating points: EER-pair $(T_1^{\text{eer}}, T_2^{\text{eer}})$, individual optimum pair $(T_1^{\text{safe}}, T_2^{\text{safe}})$, and joint optima (AND at $(T_1^*, T_2^*)_{\text{AND}}$ and OR at $(T_1^*, T_2^*)_{\text{OR}}$)

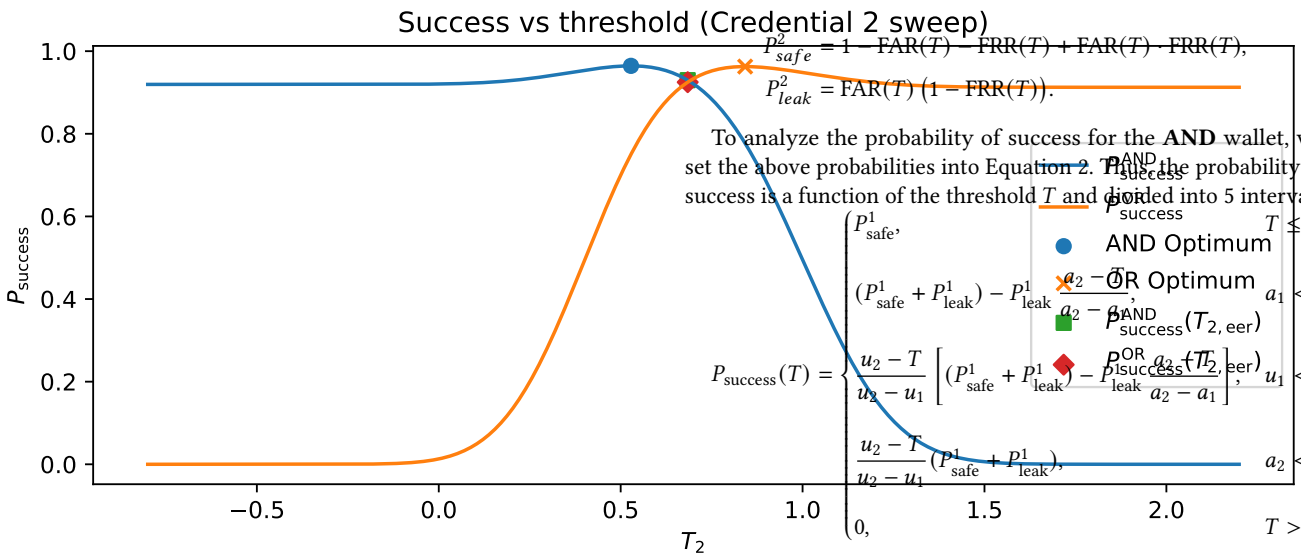


Figure 11: Success probability functions for the two Gaussian credentials

From Table 1, we observe that the optimal operating point yields a higher success probability than the EER point by between 5% to 7%.

5.2 1 Binary and 1 Biometric Credential

We started by analyzing the case of an **AND** wallet comprised of 1 binary credential and 1 biometric credential. Denote s_i the state of credential $i \in (Binary, Biometric)$, we defined the binary credential with a constant value of each one of the probabilities' states as: $P^1_{safe}, P^1_{loss}, P^1_{leak}, P^1_{theft}$ corrsponding respectively to each one of the states: safe, loss, leak and theft. The biometric credential is defined by its PDFs for the user and the attacker, P_U and P_A respectively. We firstly modeled the first order approximation of the biometric credential using uniform distribution between the bounds of u_1,u_2 corrsponding to the user distribution and a_1,a_2 corresponds to the attacker, as seen at (§4.1). We derived the probability of the biometric credential as follows:

$$P^2_{safe} = 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T) \cdot \text{FRR}(T),$$
$$P^2_{leak} = \text{FAR}(T)\,(1 - \text{FRR}(T)).$$

To analyze the probability of success for the **AND** wallet, we set the above probabilities into Equation 2. Thus, the probability of success is a function of the threshold T and divided into 5 intervals.

$$P_{\text{success}}(T) = \begin{cases} P^1_{\text{safe}}, & T \le a_1, \\ (P^1_{\text{safe}} + P^1_{\text{leak}}) - P^1_{\text{leak}}\,\dfrac{a_2 - T}{a_2 - a_1}, & a_1 < T \le u_1, \\ \dfrac{u_2 - T}{u_2 - u_1}\left[(P^1_{\text{safe}} + P^1_{\text{leak}}) - P^1_{\text{leak}}\,\dfrac{a_2 - T}{a_2 - a_1}\right], & u_1 < T \le a_2, \\ \dfrac{u_2 - T}{u_2 - u_1}(P^1_{\text{safe}} + P^1_{\text{leak}}), & a_2 < T \le u_2, \\ 0, & T > u_2 \end{cases}$$

We analyze the success probability $P_{\text{success}}(T)$ by partitioning the threshold domain into five intervals, each corresponding to a distinct relative positioning of the threshold with respect to the attacker and user distributions.

The first interval, $T \le a_1$, corresponds to the case in which the threshold is lower than the minimum value of the attacker distribution. In this regime, the FAR is equal to 1, while the FRR is equal to 0. Consequently, the safe probability of the biometric credential is zero, and the success probability converges to P^1_{safe}. Since P^1_{safe} is constant in this interval, no interior maximum exists.

The second interval, $a_1 < T \le u_1$, corresponds to the case in which the threshold lies between the minimum value of the attacker distribution and the minimum value of the user distribution. In this region, the FAR decreases linearly as a function of T, while the FRR remains equal to 0. As a result, the probabilities of the safe state and the leak state of the biometric credential are $1 - \text{FAR}(T)$ and $\text{FAR}(T)$, respectively. In this interval, $P_{\text{success}}(T)$ is a strictly increasing function, as its derivative is a constant positive value,

and therefore the maximum is attained at the boundary point

$$T_{\text{opt2}} = u_1.$$

The third interval, $u_1 < T \leq a_2$, corresponds to the case in which the threshold lies between the minimum value of the user distribution and the maximum value of the attacker distribution. In this region, both FAR and FRR vary linearly with T, resulting in a success probability that is quadratic as a function of T. Differentiating $P_{\text{success}}(T)$ yields a single critical point inside the interval,

$$T_{\text{opt3}} = \frac{u_2 + a_2 - (a_2 - a_1)\,\dfrac{P^1_{\text{safe}} + P^1_{\text{leak}}}{P^1_{\text{leak}}}}{2}$$

Since this expression is valid only for the interval $u_1 < T \leq a_2$, the effective maximizer within this interval is obtained by projecting the unconstrained optimum onto the feasible domain:

$$T^{(\text{interval})}_{\text{opt},3} = \min\big(a_2,\ \max\big(u_1,\ T_{\text{opt},3}\big)\big)\,.$$

In particular, if $T_{\text{opt},3} < u_1$, the maximum over this interval is attained at $T = u_1$, whereas if $T_{\text{opt},3} > a_2$, the maximum is attained at $T = a_2$. In all configurations consisting of one binary and one biometric credential, ingertating the wallet with the biometric credential monotonically enhances its security. Evaluating the success probability at the user threshold $T = u_1$ yields

$$P_{\text{success}}(u_1) = P^1_{\text{safe}} + P^1_{\text{leak}}(1 - \frac{a_2 - u_1}{a_2 - a_1}).$$

Since $u_1 > a_1$, it follows that

$$P_{\text{success}}(u_1) > P^1_{\text{safe}},$$

Demonstrating that the integration of the biometric credential strictly improves the security of the wallet compared to using the binary credential alone.

By examining the optimal point in this interval, we can deduce that as the variance of the attacker distribution increases, the optimal threshold T_{opt3} shifts toward the lower boundry of the interval, u_1. Moreover, for a binary credential with high safe values P^1_{safe}, and low leak values P^1_{leak}, the optimal threshold shifts toward the lower boundary as well. Thus, in such case, the success of the wallet becomes mainly dependent on the binary credential. Conversely, a binary credential with high leak values P^1_{leak} and low safe values P^1_{safe} causes the optimal threshold to shift toward the upper boundary of the interval, a_2. In this scenario, the success of the wallet relies more heavily on the biometric credential.

The fourth interval, $a_2 < T \leq u_2$, corresponds to the case in which the threshold exceeds the maximum value of the attacker distribution. In this regime, the FAR is equal to 0, while the FRR increases linearly with T. Consequently, $P_{\text{success}}(T)$ is a strictly decreasing function in this interval, and the maximum is attained at the boundary point

$$T_{\text{opt4}} = a_2.$$

Finally, the fifth interval, $T > u_2$, corresponds to the case in which the threshold exceeds the maximum value of the user and the attacker distributions, leaving the success probability equal to 0.

To determine the global optimal operating point, we compare all stationary points and boundary maxima obtained in the valid intervals. The optimal threshold is achieved in the third interval, where the attacker and user distributions overlap and the success function attains an interior maximum. Substituting $T_{\text{opt}} = T_{\text{opt3}}$ into $P_{\text{success}}(T)$ yields

$$P_{\text{success}}(T_{\text{opt}}) = \frac{\left(-a_1\left(P^1_{\text{safe}} + P^1_{\text{leak}}\right) + a_2 P^1_{\text{safe}} + P^1_{\text{leak}}\,u_2\right)^2}{4P^1_{\text{leak}}(a_2 - a_1)(u_2 - u_1)}.$$

We have done similar analysis for the **OR** wallet. We defined the credentials' probabilities of the wallet as follows:

$$\begin{aligned}
P_{\text{safe1}} &= 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T)\cdot\text{FRR}(T),\\
P_{\text{safe2}} &= k_1,\\
P_{\text{loss1}} &= \text{FRR}(T)\,\big(1 - \text{FAR}(T)\big),\\
P_{\text{loss2}} &= k_2.
\end{aligned}$$

Therefore, the probability of success for the **OR** wallet is defined as:

$$P_{\text{success}}(T) = \begin{cases}
0, & T \leq a_1,\\
\dfrac{T - a_1}{a_2 - a_1}\,(k_1 + k_2), & a_1 < T \leq u_1,\\
\dfrac{T - a_1}{a_2 - a_1}\left[k_1 + k_2 - k_2\,\dfrac{T - u_1}{u_2 - u_1}\right], & u_1 < T \leq a_2,\\
k_1 + k_2 - k_2\,\dfrac{T - u_1}{u_2 - u_1}, & a_2 < T \leq u_2,\\
k_1, & T > u_2.
\end{cases}$$

By differentiating every interval and identifying the critical points, we found that the optimal threshold that leads to the maximum probability of success lies in the third interval where $u_1 < T \leq a_2$ and is given by:

$$T_{\text{opt}} = \frac{a_1 k_2 + u_2(k_1 + k_2) - k_1 u_1}{2k_2}$$

And the maximum probability of success at this point is:

$$P_{\text{success}}(T_{\text{opt}}) = \frac{\left(a_1 k_2 - u_2(k_1 + k_2) + k_1 u_1\right)^2}{4k_2(a_2 - a_1)(u_2 - u_1)}$$

For both wallet types, we observed that the optimal operating point T_{opt} differs from the equal error rate (EER) point T_{EER}. As the EER depends on the PDF's type solely, we calculated the EER point and showed each wallet's success in Figure 12 and Figure 13 respectively.

$$T_{\text{EER}} = \frac{a_1 u_1 - a_2 u_2}{a_1 - a_2 + u_1 - u_2}$$

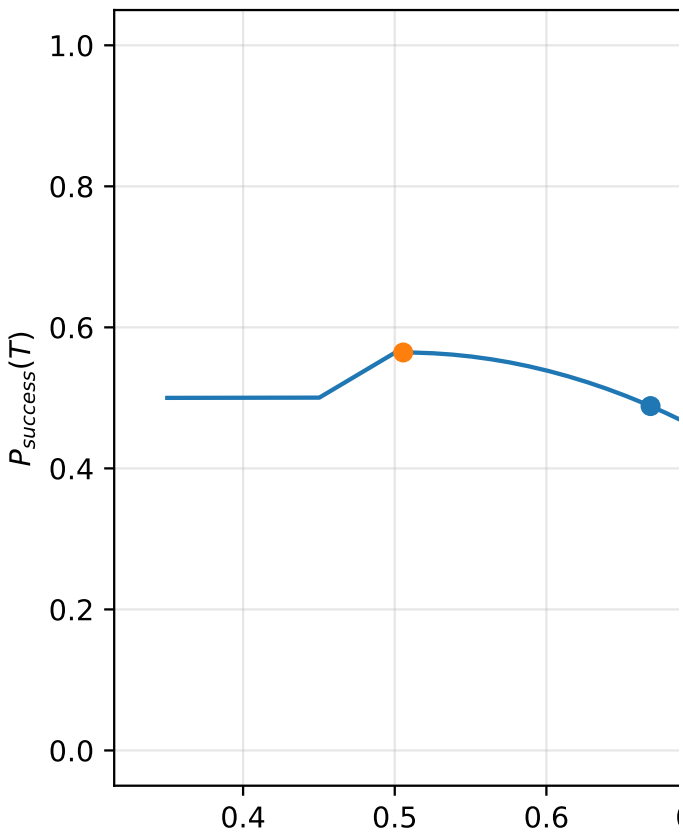


Figure 12: Success probability functions for one binary and one biometric credential in an AND wallet

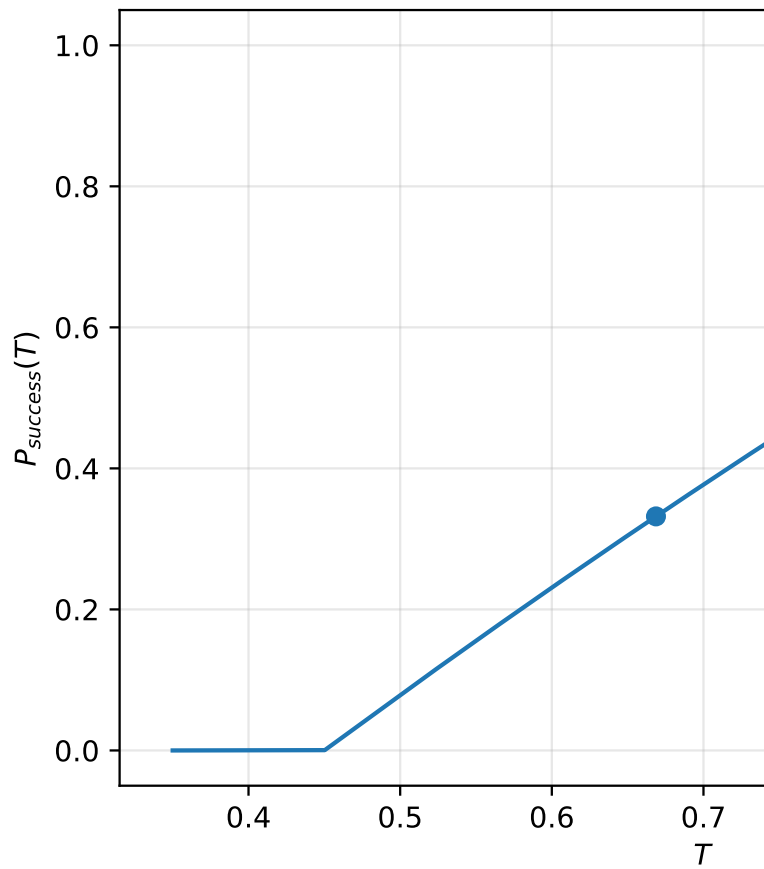


Figure 13: Success probability functions for one binary and one biometric credential in an OR wallet

6 Characterizing Security Improvement

We will now characterize the conditions under which integrating a biometric credential into a wallet composed of a single binary credential can enhance its security. First, we show that for any wallet whose binary credential characterized by non-zero probabilities P_{safe} and P_{theft} only, such integration cannot improve security (§6.1). Next, we present a necessary and sufficient condition establishing that, for any wallet composed of a binary credential with non zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss}, integrating a biometric credential improves security for both finite and infinite distributions (§6.2). Finally, we provide a more intuitive sufficient condition under which a biometric credential enhances the security of a wallet based on a single binary credential with non zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss} (§6.3).

6.1 No Security Gain in the Safe/Theft Case

Proposition 1. *For any wallet w composed of a single binary credential with non-zero probabilities P_{safe} and P_{theft} only, integrating the wallet with a biometric credential does not enhance its security.*

Proof. Consider a wallet w composed of a single binary credential with non-zero probabilities P_{safe} and P_{theft} only. In this case, integrating the wallet with a biometric credential lead to success probability described by Equation 2 and Equation 3. Since $P_{\text{leak}} = P_{\text{loss}} = 0$, the success probability in both wallet cases reduces to:

$$P_{\text{success}}(T) = P^1_{\text{safe}} \cdot P^2_{\text{safe}}(T)$$

Since $P^2_{\text{safe}}(T) \leq 1$ for every value of T, it follows that $P_{\text{success}}(T) \leq P^1_{\text{safe}}$, and thus integrating a biometric credential does not enhance the security of the wallet. □

6.2 Sufficient And Necessary Condition for Integrating Biometric Credentials

First, we will define the notion of *asymptotic separability* of a biometric credential. Then, we will show that its a necessary and sufficient condition for enhancing the security for both finite and infinite distributions.

Definition 1. A biometric credential with user distribution function $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically separable* if

$$\inf_t \min\left\{\frac{\text{FAR}(t)}{1-\text{FRR}(t)}, \frac{\text{FRR}(t)}{1-\text{FAR}(t)}\right\} = 0,$$

where $\text{FAR}(t) = \int_t^{\infty} P_a(s)ds$ and $\text{FRR}(t) = \int_{-\infty}^{t} P_u(s)ds$.

Proposition 2. *For any wallet w composed of a single binary credential with non-zero probabilities P_{safe}, P_{leak} or P_{loss}, integrating the wallet with a biometric credential whose user and attacker distributions are asymptotically separable, enhances the security of the wallet.*

Proof. The security of a wallet composed of a single binary credential with non-zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss} is given by P^1_{safe}. A biometric credential is characterized by 4 probability states as function of T: $P^2_{\text{safe}}(T), P^2_{\text{loss}}(T), P^2_{\text{leak}}(T), P^2_{\text{theft}}(T)$. We can describe each one of these probabilities as a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$\begin{aligned}
P^2_{\text{safe}}(T) &= 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T)\text{FRR}(T),\\
P^2_{\text{loss}}(T) &= \text{FRR}(T)\big(1 - \text{FAR}(T)\big),\\
P^2_{\text{leak}}(T) &= \text{FAR}(T)\big(1 - \text{FRR}(T)\big),\\
P^2_{\text{theft}}(T) &= \text{FAR}(T)\text{FRR}(T).
\end{aligned}$$

The success probability of the integrated wallet comprised of 2 credentials is described by Equation 2 and Equation 3. Thus, by placing the biometric credential in each configuration we can express the success probability of the integrated wallet as a function of the operating threshold T:

$$P^{\text{AND}}_{\text{success}}(T) = \big(1 - \text{FRR}(T)\big)\Big(P^1_{\text{safe}} + P^1_{\text{leak}}\big(1 - \text{FAR}(T)\big)\Big). \quad (4)$$

$$P^{\text{OR}}_{\text{success}}(T) = \big(1 - \text{FAR}(T)\big)\Big(P^1_{\text{safe}} + P^1_{\text{loss}}\big(1 - \text{FRR}(T)\big)\Big). \quad (5)$$

We need to show that there exists an operating threshold T such that either $P^{\text{AND}}_{\text{success}}(T) > P^1_{\text{safe}}$ or $P^{\text{OR}}_{\text{success}}(T) > P^1_{\text{safe}}$.

By algebraic manipulation of Equation 4 we can demand $P^{\text{AND}}_{\text{success}}(T) > P^1_{\text{safe}}$ if and only if:

$$\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}} \cdot (1 - \text{FAR}(T))}$$

Since $P^1_{\text{leak}} \cdot (1 - \text{FAR}(T)) < P^1_{\text{leak}}$, a sufficient condition for the above inequality is:

$$\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}}} \quad (6)$$

Similarly, we can deduce that $P^{\text{OR}}_{\text{success}}(T) > P^1_{\text{safe}}$ if and only if:

$$\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}} \cdot (1 - \text{FRR}(T))}$$

Since $P^1_{\text{loss}} \cdot (1 - \text{FRR}(T)) < P^1_{\text{loss}}$, a sufficient condition for the above inequality is:

$$\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}}} \quad (7)$$

By the definition of asymptotic separability, there exists an operating threshold T such that:

$$\inf_t \min\left\{\frac{\text{FAR}(t)}{1-\text{FRR}(t)}, \frac{\text{FRR}(t)}{1-\text{FAR}(t)}\right\} = 0.$$

Equivalently, $\forall \varepsilon > 0, \exists T$ such that either $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < \varepsilon$ or $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < \varepsilon$. Thus, we can choose T such that ε satisfies either Equation 6 or Equation 7. More precisely, if

$$\inf_T \frac{\text{FAR}(T)}{1-\text{FRR}(T)} = 0,$$

then the OR wallet composition is preferable, whereas if

$$\inf_T \frac{\text{FRR}(T)}{1-\text{FAR}(T)} = 0,$$

then the AND wallet composition is preferable. Therefore, integrating an asymptotically separable biometric credential enhances the security of the wallet.

□

Proposition 3. *Suppose there $\exists T$ such that the integrated wallet w composed of biometric and binary credential with non zero probabilities P_{safe}, P_{leak} or P_{loss} achieves a strictly higher success probability than the binary-only wallet, i.e.,*

$$P^{integrated}_{success}(T) > P^{binary}_{success}.$$

Then necessarily, the biometric credential is asymptotically separable.

Proof. We prove by contradiction. Assume for the sake of contradiction, that the biometric credential is not asymptotically separable. Then $\exists \varepsilon > 0$ such that $\forall T$ either $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} \geq \varepsilon$ or $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} \geq \varepsilon$. Since it's given that the integrated wallet w achieves a strictly higher success probability than the binary-only wallet, then there exists an operating threshold T such that either $P^{\text{AND}}_{\text{success}}(T) > P^1_{\text{safe}}$ or $P^{\text{OR}}_{\text{success}}(T) > P^1_{\text{safe}}$. Thus, either Equation 6 or Equation 7 holds for every value of $P^1_{\text{safe}}, P^1_{\text{leak}}, P^1_{\text{loss}}$ in the range of $(0, 1)$. In particular, we can choose $P^1_{\text{safe}}, P^1_{\text{leak}}, P^1_{\text{loss}}$ such that $\forall \varepsilon > 0 : \frac{P^1_{\text{leak}}}{P^1_{\text{safe}}+P^1_{\text{leak}}} < \varepsilon$ and $\frac{P^1_{\text{loss}}}{P^1_{\text{safe}}+P^1_{\text{loss}}} < \varepsilon$. Thus, we can deduce that both $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} \geq \frac{P^1_{\text{leak}}}{P^1_{\text{safe}}+P^1_{\text{leak}}}$

and $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} \geq \frac{P^1_{\text{loss}}}{P^1_{\text{safe}}+P^1_{\text{loss}}}$, which contradicts the assumption that the integrated wallet achieves a strictly higher success probability than the binary-only wallet. Therefore, the biometric credential must be asymptotically separable.

□

We showed that *asymptotic separability* of the user and attacker distributions is a sufficient and necessary condition for enhancing the security of a wallet composed of a single binary credential with non zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss} by integrating it with a biometric credential.

6.3 Intuitive Sufficient Condition

Definition 2. A biometric credential with user distribution fucntion $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically dominated* if:

$$\forall c > 0 \ \exists T \text{ s.t.} : \forall x \geq T : p_a(x) \leq c\, p_u(x)$$

or

$$\forall c > 0 \ \exists T \text{ s.t.} : \forall x \leq T : p_u(x) \leq c\, p_a(x)$$

We will show that *asymptotically dominancy* of user and attacker distribution functions is a sufficient condition for enhancing a binary based wallet's security.

Proposition 4. *For any wallet w composed of one binary credential and success function $P^{binary}_{success}$, augmenting the wallet with an asymptotically dominated biometric credential improves its achievable security. That is,*

$$P^{integrated}_{success}(T) > P^{binary}_{success}.$$

Proof. A binary credential is characterized by fixed probability states $P^1_{\text{safe}}, P^1_{\text{loss}}, P^1_{\text{leak}}, P^1_{\text{theft}}$. Thus the success probability of a wallet composed solely of this binary credential is constant and given by:

$$P^{\text{binary}}_{\text{success}} = P^1_{\text{safe}}$$

A biometric credential is characterized by 4 probability states that depend on the operating threshold T: $P^2_{\text{safe}}(T), P^2_{\text{loss}}(T), P^2_{\text{leak}}(T), P^2_{\text{theft}}(T)$. We can describe each one of these probabilities as a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$\begin{aligned}
P^2_{\text{safe}}(T) &= 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T)\text{FRR}(T),\\
P^2_{\text{loss}}(T) &= \text{FRR}(T)\big(1 - \text{FAR}(T)\big),\\
P^2_{\text{leak}}(T) &= \text{FAR}(T)\big(1 - \text{FRR}(T)\big),\\
P^2_{\text{theft}}(T) &= \text{FAR}(T)\text{FRR}(T).
\end{aligned}$$

The success probability of the integrated wallet comprised of 2 credentials is described by Equation 2 and Equation 3. Thus, we can express the success probability of the integrated wallet as a function of the operating threshold T as shown in Equation 4 and Equation 5. We previously observed that a sufficient condition for proving improvement of achievable security is described by Equation 6 and Equation 7. We need to show that there exists an operating threshold T such that either:

$$\frac{\text{FRR}(T)}{1 - \text{FAR}(T)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}}}$$

or

$$\frac{\text{FAR}(T)}{1 - \text{FRR}(T)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}}}$$

We will expand left hand side of Equation 6.

$$\frac{\text{FRR}(T)}{1 - \text{FAR}(T)} = \frac{\int_{-\infty}^{T} P_u(s)ds}{\int_{-\infty}^{T} P_a(s)ds}$$

By Definition 2, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold T such that for every $s \leq T$, $P_u(s) \leq c \cdot P_a(s)$. Therefore:

$$\frac{\int_{-\infty}^{T} P_u(s)ds}{\int_{-\infty}^{T} P_a(s)ds} \leq \frac{\int_{-\infty}^{T} c \cdot P_a(s)ds}{\int_{-\infty}^{T} P_a(s)ds} = c$$

This is true $\forall c > 0$, thus we can choose c such that it satisfies Equation 6.
We will expand left hand side of Equation 7.

$$\frac{\text{FAR}(T)}{1 - \text{FRR}(T)} = \frac{\int_{T}^{\infty} P_a(s)ds}{\int_{T}^{\infty} P_u(s)ds}$$

By Definition 2, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold T such that for every $s \geq T$, $P_a(s) \leq c \cdot P_u(s)$. Therefore:

$$\frac{\int_{T}^{\infty} P_a(s)ds}{\int_{T}^{\infty} P_u(s)ds} \leq \frac{\int_{T}^{\infty} c \cdot P_u(s)ds}{\int_{T}^{\infty} P_u(s)ds} = c, \forall c > 0$$

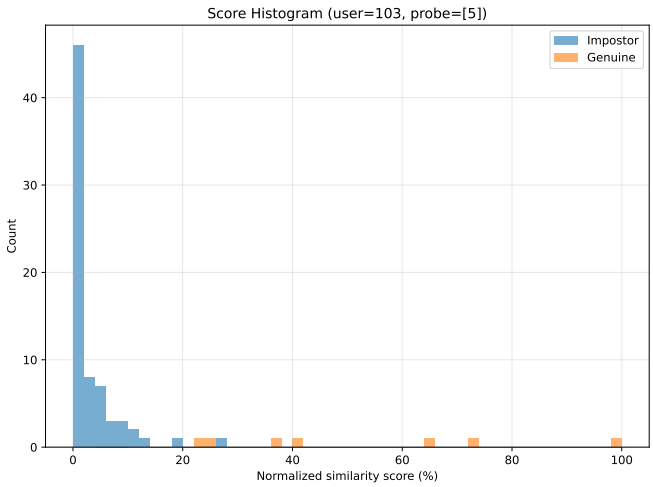
We showed that $\forall c > 0$ either $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < c$ or $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < c$. Therefore, integrating a binary based wallet with an asymptotically dominated biometric credential improves its security. □

This proof is not trivial, as the integration of another binary credential to a binary credential based wallet, does not always enhance its security. For example, if the wallet is comprised of 1 binary credential with high safe values and low leak values, adding a binary credential with low safe values and high leak values would decrease the overall security. Thus, Proposition 4 shows that by using an *asymptotically dominated* biometric credential, we can always improve the security of any binary credential based wallet. Moreover, *asymptotically dominancy* condition supports practicale scenarios, as we expect that for very low values of score s the attacker distribution dominates the user while vise versa for very high values of s. We will show measuremens supports this claim in Section 6.

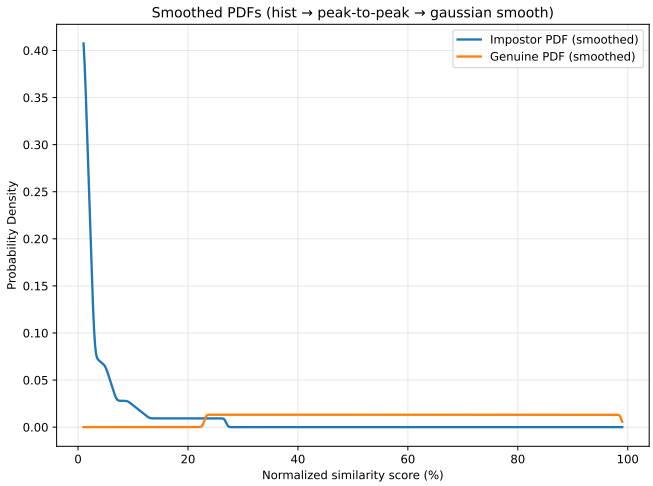
7 Measuremens

We will now present measurements based on fingerprint taken from FVC2002 database [1] and FVC2004 database [2]. We used SourceAFIS an opensource algorithm created by Vazan [24] for scoring fingerprints out of their images based on minutiae triplets. We chose to use fingerprints, as it is a widely common and used example of a biometric credential in authentication mechanisms. For a test set, we took a template out of FVC2002 database 1 of user 103 and compare them against the whole database of both FVC2002 and FVC2004. A geniune will be considered the template of user 103 against its other templates while imposter will be all the other combinations. We maintained same approach as Jain et al [9] and Cappelli et al [5] and first built histogram of the geniune and

imposter combinations as shown in Figure 14a. Then, we turned the histogram into a smoothed probability distribution function as shown in Figure 14b. We found the False rejection rate and False acceptance rate and computed the Equal Error Rate (EER) as shown in Figure 15. The most important outcome from this approach was to achieve similar EER rate of around 2% as achieved from Maio et al. [12] so we could objectively make a comparison. We computed the probability success function as given in Equation 1 of a wallet composed of a single biometric credential and showed that the optimal operating point is different from the EER point and achieves higher success probability as shown in Figure 16a. Finally, we computed the probability success function of a wallet composed of a single binary credential with $P_{\text{safe}} = 0.9$ and $P_{\text{leak}} = 0.04$ $P_{\text{loss}} = 0.04$ and integrated it with the biometric credential.



(a) Histogram of the genuine and imposter combinations of the fingerprint scores



(b) Smoothed probability distribution function of the genuine and imposter combinations of the fingerprint scores

Figure 14: From histogram to probability distribution function

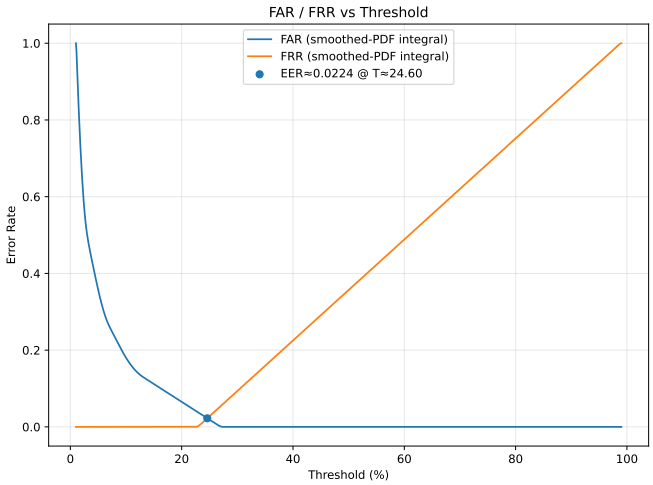
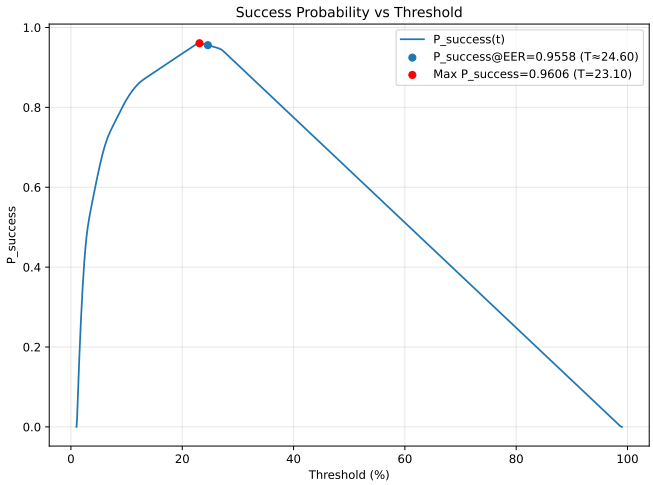


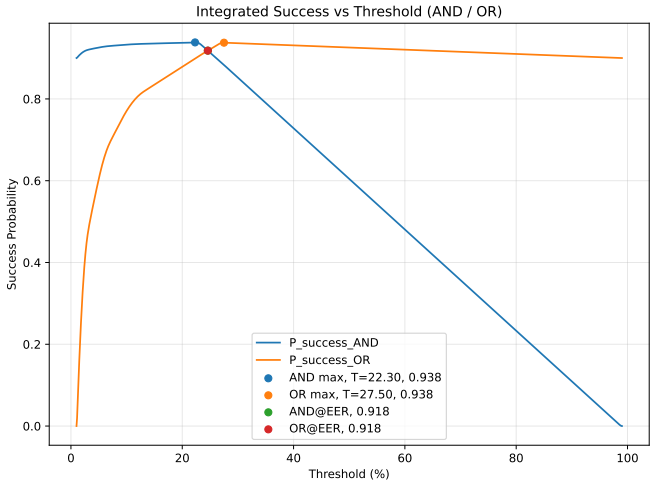
Figure 15: FAR vs. FRR curves for the fingerprint scores

We showed that the optimal operating point of the integrated wallet is different from the EER point and achieves higher success probability by around 1.6% than the EER point, as the EER itself is only 1.8%. In addition, we can observe that the optimal point of the wallet achieves higher success probability than both the discrete only and continuous only wallets as shown in Figure 16b. Although there is an improvement in the success probability of the integrated wallet, we can observe the improvement is not as significant as the one observed in the theoretical analysis. This can be explained by the fact that the EER point of the biometric credential is already very low, thus there is not much room for improvement. In that scenario, we compared several templates of user 103 against templates of different users. However, our main approach in this paper was to increase the security level and success in scenarios where an imposter will try to spoof the user templates and authenticate into the user account. Hence, we expect the imposter to use spoofing tools, producing distribution function that much more similar to the genuine distribution function, which might lead to increased EER point.

To support this claim, we used the LivDet2015 database [7, 14, 17, 25], which contains a large-scale dataset of live and spoof fingerprint samples. We followed the same methodology as before and used the SourceAFIS algorithm [24]. As probe templates, we selected six Digital Persona scanner images of user 002_0. Each probe was matched against 14 additional images of the same user to generate genuine scores. For each target image, we define the final score as the maximum similarity across all probe templates. For example, if the matching scores between a target image and the six probes are [0.3, 0.5, 0.4, 0.8, 0.9, 0.7], then the resulting score is 0.9. To simulate a spoofing scenario, we used fake fingerprint images of the same user (002_0) produced using gelatine and matched them against the probe templates to obtain the imposter score distribution. The same maximum-score aggregation was applied in this case as well. We present the resulting histogram and the smoothed probability distribution functions in Figures 17a and 17b. We then computed the EER point and plotted the FAR-FRR curve, as shown

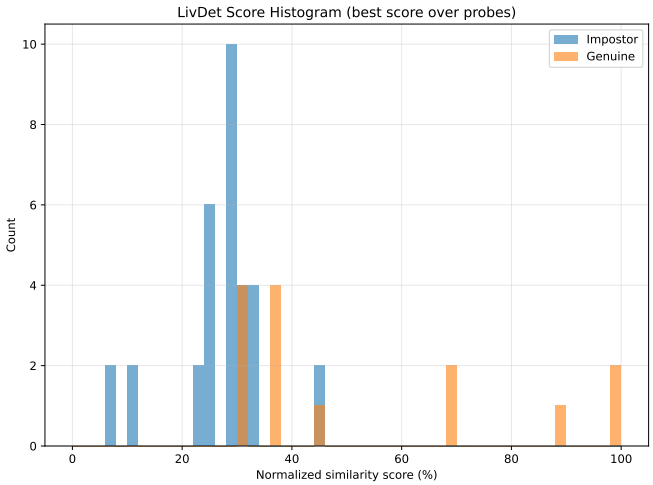


(a) Success probability function of a wallet composed of a single biometric credential

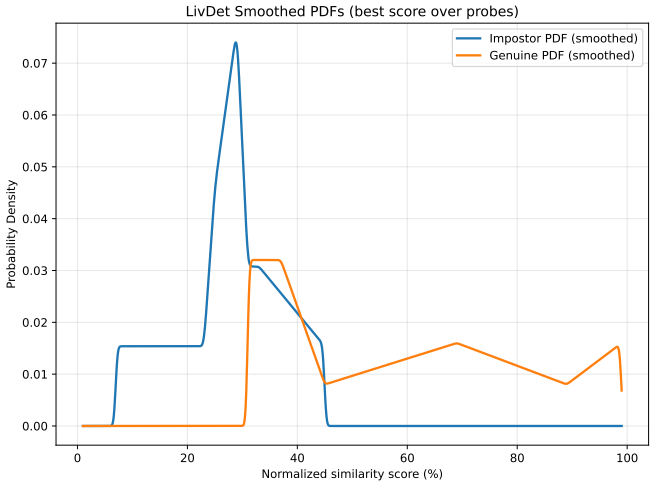


(b) Success probability function of a wallet composed of a single binary credential and a single biometric credential

Figure 16: Biometric credential achieves higher success probability than the EER point in both biometric credential based wallet and integrated wallet



(a) Histogram of the genuine and imposter combinations of the fingerprint scores



(b) Smoothed probability distribution function of the genuine and imposter combinations of the fingerprint scores

Figure 17: From histogram to probability distribution function

in Figure 18. As expected, the EER increased significantly to 17.8%. Finally, we evaluated the success probability for both a single biometric credential, as shown in Figure 19a, and for a wallet composed of a single binary credential and a single biometric credential, as shown in Figure 19b. We can observe that the optimal operating point achieves a success probability approximately 14% higher than the EER point when integrating it with a binary credential with $P_{\text{safe}} = 0.8, P_{\text{loss}} = 0.04, P_{\text{leak}} = 0.15, P_{\text{theft}} = 0.01$ parameters, using the same algorithm.

8 Conclusion

We formalize the wallet authentication model consisting, for the first time, of both biometric alone and multi-credentials wallets. We analyze the security of each wallet type and present a new operating point that achieves higher success probability by modelling each credential with four probability states. We show every wallet comprised of binary credentials can be enhanced by integrating it with asymptotically separable biometric credential. Our analysis shows significant improvement of the success probability under spoofing attack scenarios, which became prevalant and easier on biometric systems in today's artificial intelligence era. Building new mechanisms, based on the suggested dynamic operating point done

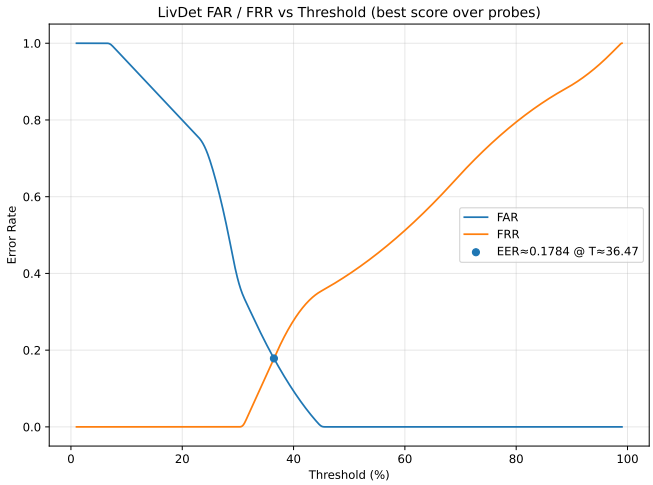
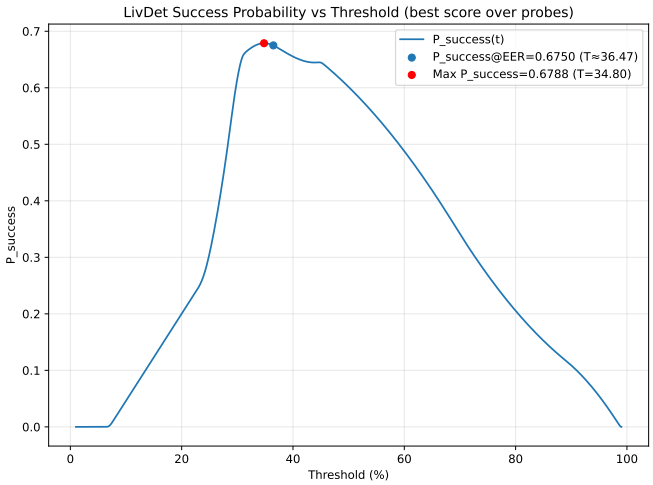
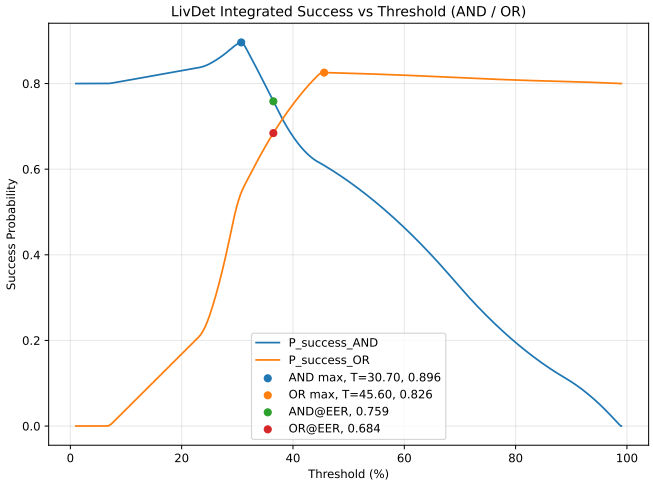


Figure 18: FAR vs. FRR curves for the fingerprint scores



(a) Success probability function of a wallet composed of a single biometric credential



(b) Success probability function of a wallet composed of a single biometric credential and a single binary credential with probabilities $P_{\mathbf{safe}} = 0.8$, $P_{\mathbf{loss}} = 0.04$, $P_{\mathbf{leak}} = 0.15$, $P_{\mathbf{theft}} = 0.01$.

Figure 19: Biometric credential achieves higher success probability than the EER point in both biometric credential based wallet and integrated wallet

(§4) and stricting the threshold when adversaries are using spoofing tools to attack the biometric system is a promising direction for future work. Our findings have important and impactful implications for the design of secure authentication biometric systems. Harnessing our approach, will affect millions of users around the world and improve their privacy and security of their digital assets.

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